Coding Challenge

Write a program that analyzes a CSV-like table and generates a data quality report showing the count of missing values and the percentage of valid (non-missing) values for each column.

Requirements

1. Read a CSV table with header row and data rows
2. For each column, calculate:
    - **Missing count:** Number of missing values
    - **Valid percentage:** Percentage of non-missing values (rounded to 2 decimals)
3. Identify missing values:
    - Empty string: **""**
    - The literal text: **"NULL"** (case-sensitive)
    - Missing trailing fields in short rows
4. Output format: **column:missing:valid_percentage** for each column
5. Handle edge cases: blank fields, short rows, fully missing columns

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Missing Value Rules

What Counts as Missing?

Value Type	Example	Is Missing?
Empty string	**,,** or **,** at end	✓ Yes
Literal "NULL"	**NULL** (exact text)	✓ Yes
Short row	Row has fewer fields than header	✓ Yes (trailing fields)
Whitespace only	" " or " "	✗ No (valid value)
Zero	**0**	✗ No (valid value)
Text "null"	**null** (lowercase)	✗ No (valid value)
Number	**123** or **-45**	✗ No (valid value)

Important Notes

- **Case-sensitive:** Only **NULL** (uppercase) is missing
- **Trailing fields:** If a row has fewer commas, trailing fields are missing
- **Empty vs NULL:** Both count as missing

Input Format

- **Line 1:** Header row (comma-separated column names)
- **Lines 2+:** Data rows (comma-separated values)
- No quoted commas (simple CSV parsing)
- Maximum: **100 columns**, **100,000 rows**

Output Format

For each column (in header order):

column_name:missing_count:valid_percentage

- **column_name:** From header row
- **missing_count:** Integer count of missing values
- **valid_percentage:** Percentage of valid values, rounded to **exactly 2 decimal places**

Valid Percentage Calculation

valid_percentage = (valid_count / total_rows) × 100

Where:

- valid_count = total_rows - missing_count

- total_rows = number of data rows (excluding header)

Examples

Example 1: Basic Missing Values

Input:

id,name,score

1,Ana,80

2,,90

3,Cia,NULL

Output:

id:0:100.00

name:1:66.67

score:1:66.67

Explanation:

- **Total data rows:** 3
- **id column:**
  - Row 1: **1** ✓ valid
  - Row 2: **2** ✓ valid
  - Row 3: **3** ✓ valid
  - Missing: 0, Valid: 3/3 = 100.00%
- **name column:**
  - Row 1: **Ana** ✓ valid
  - Row 2: `` ` `` ✗ empty (missing)
  - Row 3: **Cia** ✓ valid
  - Missing: 1, Valid: 2/3 = 66.67%
- **score column:**
  - Row 1: **80** ✓ valid
  - Row 2: **90** ✓ valid
  - Row 3: **NULL** ✗ missing
  - Missing: 1, Valid: 2/3 = 66.67%

Example 2: Short Row (Missing Trailing Fields)

Input:

a,b,c

1,2,3

4,5

6,7,8

Output:

a:0:100.00

b:0:100.00

c:1:66.67

Explanation:

- **Total data rows:** 3
- **Header has 3 columns:** a, b, c
- **Row 2 is short: 4,5** (missing column c)
- **Column a:** All present: [1, 4, 6] → Missing: 0
- **Column b:** All present: [2, 5, 7] → Missing: 0
- **Column c:** [3, missing, 8] → Missing: 1, Valid: 2/3 = 66.67%